

Worker Adjustment, Skills, Training, And Career Transitions

Supply-Side Adjustment Under Technological Change

Special Topic 5: Technology, Innovation, and Labor – Week 3

Central question

- When technology changes task prices and career risk, which workers can adjust?
- Which workers adapt through skills, training, mobility, or job redesign?
- Which workers bear persistent earnings, employment, and welfare losses?
- Week 3 discipline: aggregate reallocation is not the same object as worker welfare.

From demand shocks to supply response

- Week 2: automation and AI shift labor demand across tasks, firms, occupations, and places.
- Week 3: exposed workers choose whether to learn, retrain, search, switch, move, bargain, or exit.
- The labor object is the cost and feasibility of adjustment.
- The policy object is not generic reskilling; it is incidence across workers with different constraints.

Worker-adjustment framework

$$V_{it} = w_{it}(s_{it}, o_{it}, f_{it}, \tau_t) - c_i(a_{it}) - \kappa_i(m_{it}) + E_t \sum_{r=t+1}^T \beta^{r-t} u_{ir}$$

- s : skills; o : occupation; f : employer; τ : technology environment.
- a : costly adjustment effort such as training, coursework, search, or learning.
- m : mobility across employers, occupations, sectors, or places.
- Technology changes current pay, future skill values, and the option value of moving.

Adjustment margins

Margin	Labor object	Main question
Early skill choice	major, field, vocational track	who enters resilient skill paths?
Within-firm adaptation	retraining, reassignment, learning	do incumbents adapt or get replaced?
Between-job mobility	employer, occupation, sector, location	can workers move to growing tasks?
Labor-force exit	unemployment, disability, retirement	is exit substituting for re-training?

- Staying can mean adaptation or being stuck; moving can mean opportunity or displacement.

Aggregate adjustment vs worker losses

- A market can fill new vacancies while original exposed workers lose earnings for years.
- Firms can upgrade skill composition by hiring new workers instead of retraining incumbents.
- Average wages can rise through composition even when displaced workers are worse off.
- Local employment can recover through entry, migration, or retirement of exposed workers.
- The welfare question follows workers, not only jobs.

Skill specificity and portability

- Portable skills retain value across firms, occupations, technologies, and local labor markets.
- Specific skills can pay well before a shock and depreciate sharply after tasks change.
- STEM and other technology-intensive careers can require repeated updating as tools change.
- Two equally exposed workers can face different losses if one can redeploy skills and the other cannot.

- Majors, fields, vocational tracks, and first occupations shape future task exposure.
- The main empirical challenge is selection into fields and training paths.
- Cutoff and centralized-admissions designs compare students near assignment thresholds.
- These designs are strongest for access to early skill pathways, not for all later-career displacement.

Training and retraining

- Training can be formal, informal, employer-provided, public, worker-financed, or embedded in job redesign.
- Take-up is selected: time, information, motivation, liquidity, and employer support all matter.
- A useful design measures take-up, completion, earnings, employment, job quality, and persistence.
- Training can reduce inequality if it reaches exposed workers; it can widen inequality if advantaged workers capture it.

Within-firm adaptation

- Technology is often implemented through managers, software systems, workflows, and internal labor markets.
- Matched employer-worker data can track incumbents before and after adoption.
- Key outcomes: retention, exit, earnings growth, task reassignment, occupation switching, and hiring composition.
- Main threat: adopters and their workers may already be on different trajectories.

Methods and design

Design	Identifies well	Main caution
Major/track cutoff RD	returns to skill pathways	local to cutoff
Training evaluation	causal effect of offer or access	take-up selection
Matched worker-firm event study	adaptation around adoption	adopter selection
Local exposure + admin data	equilibrium training or retirement response	place-level incidence
Worker patent/task exposure	worker-level technology incidence	exposure mapping
Firm AI/task panels	adoption, training, and job redesign	short horizons

What a strong design states

- ① Technology shock: robot, AI, software, patent, digital tool, or task automation.
- ② Exposed workers: occupation, firm, industry, place, field, task bundle, or skill history.
- ③ Adjustment margin: training, switching, reassignment, migration, retirement, or nonemployment.
- ④ Outcome horizon: immediate, medium-run, or career.
- ⑤ Counterfactual and welfare object: earnings, employment, job quality, risk, or broader welfare.

Worker frictions

Friction	Adjustment problem	Inequality channel
Distrust / skepticism	avoid new tools or training providers	past experience differs
Inertia / present bias	delay costly learning with future returns	delays are costly
Information frictions	do not know which skills pay	networks matter
Liquidity and time costs	cannot finance training or search	wealth matters
Switching frictions	credentials, housing, family, identity	mobility differs

- Frictions affect both take-up and the distribution of technology gains.

$$Loss_i(H) = \sum_{h=0}^H \beta^h [Y_{i,t+h}^0 - Y_{i,t+h}^1]$$

- Y can be earnings, employment, hours, amenities, or job quality.
- Fast reemployment can still leave a long earnings loss.
- Remaining employed can still mean lower autonomy, lower advancement, or weaker bargaining.
- The horizon matters because adjustment costs are dynamic.

- Reproduce: construct synthetic worker-level technology exposure from task shares.
- Diagnose: classify adjustment as staying, training, switching, unemployment, or exit.
- Transfer: compare training and retirement responses under local exposure.
- Challenge transfer: compare incumbent training and new-hire training under AI adoption.
- Goal: see why aggregate skill upgrading can hide incumbent worker losses.

- Week 3 treats workers as the unit that must adjust.
- Week 4 moves inside firms.
- The next question is how adoption, hiring, management, internal labor markets, and job redesign shape worker adjustment.

- Technology changes the return to skills and the risk attached to careers.
- Early choices, within-firm retraining, mobility, and exit are separate margins.
- Strong empirical designs measure dynamic worker outcomes and confront selection.
- Frictions can turn a productivity-enhancing technology into unequal worker incidence.